



9. Proactive Cybersecurity

Jin Hong

jin.hong@uwa.edu.au

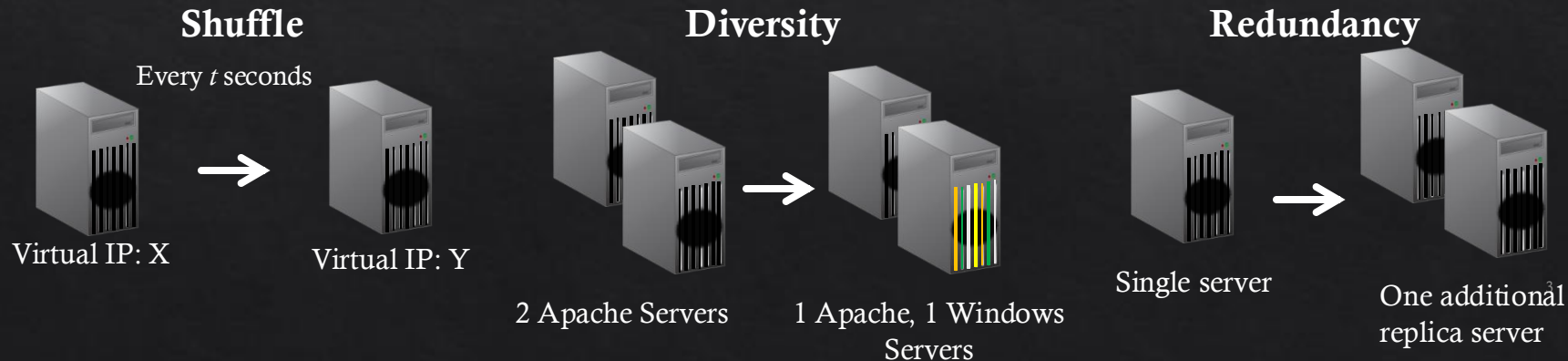
Emerging Security Issues: outline

- ◇ Emerging Security Issues
 - ◇ Emerging networks
 - ◇ cloud, SDN, IoT
 - ◇ Vehicle security
 - ◇ Homomorphic encryption
 - ◇ Security and Humans

Moving Target Defense



- ◇ Simply MTD, is a defense mechanism that continuously changes the attack surface to thwart cyberattacks
- ◇ Different types of changes can be made in the system, and they can be categorised into three



MTD

- ◇ MTD can also be implemented in different layers of the system
 - ◇ For example:
 - ◇ Application layer – code regeneration/diversification
 - ◇ Transport layer – routing node redundancy
 - ◇ Internet layer – IP shuffling and virtualisation

MTD

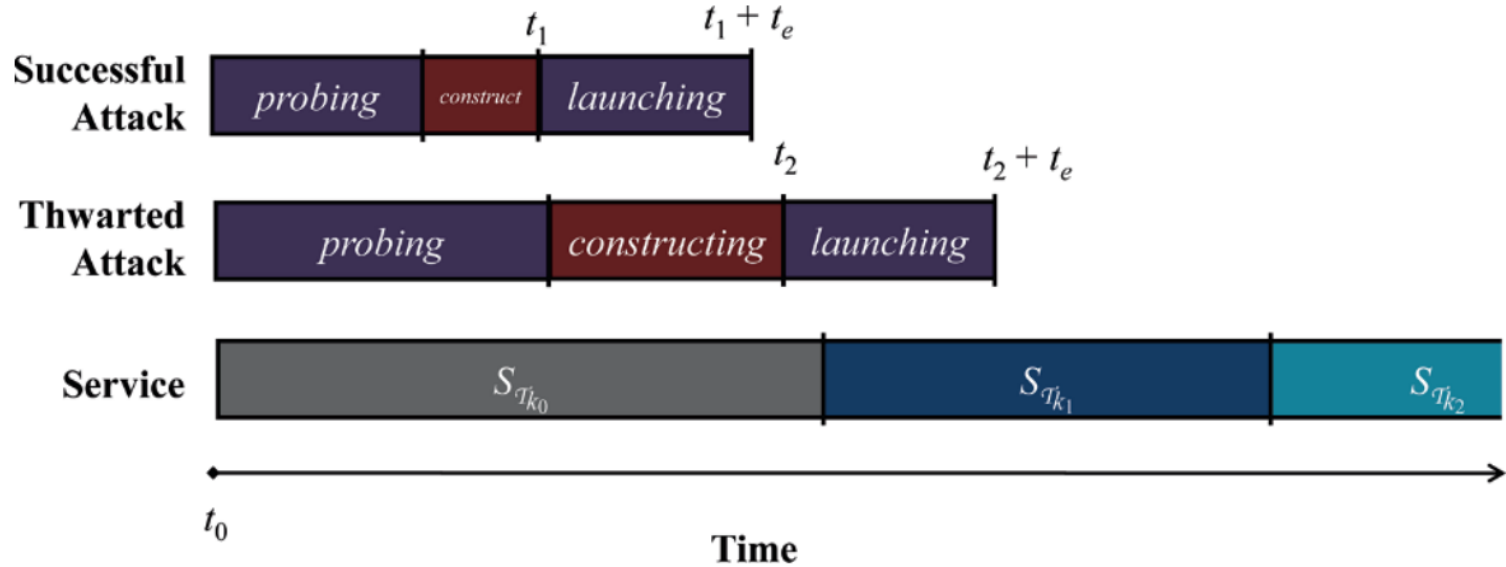


Fig. 2.1 Attack Lifetime

MTD Shuffle

- ◇ The MTD shuffle technique **changes** the configuration of the system such that connections/dependencies are altered but the operations are maintained
 - ◇ E.g., changing the routing table in the SDN
- ◇ Existing vulnerabilities are **not removed**, but it redirects the attack sequence to be taken
 - ◇ Hence, delaying the reconnaissance process, as well as breaking an ongoing attack

MTD Shuffle

- ◇ Example: Openflow Random Host Mutation
 - ◇ Typically, a host has one (physically) IP address
 - ◇ In HF-RHM scheme, each host has a physical IP address and a virtual IP address
 - ◇ Only the SDN controller knows the physical IP address of each host
 - ◇ Virtual IP (vIP) addresses are used for establishing communication channels
 - ◇ vIP addresses are shuffled every x seconds, OF switches are updated to redirect flows and update the existing communications
 - ◇ Attackers probing for IP addresses loses the identified victim host as vIP shuffles

MTD Shuffle

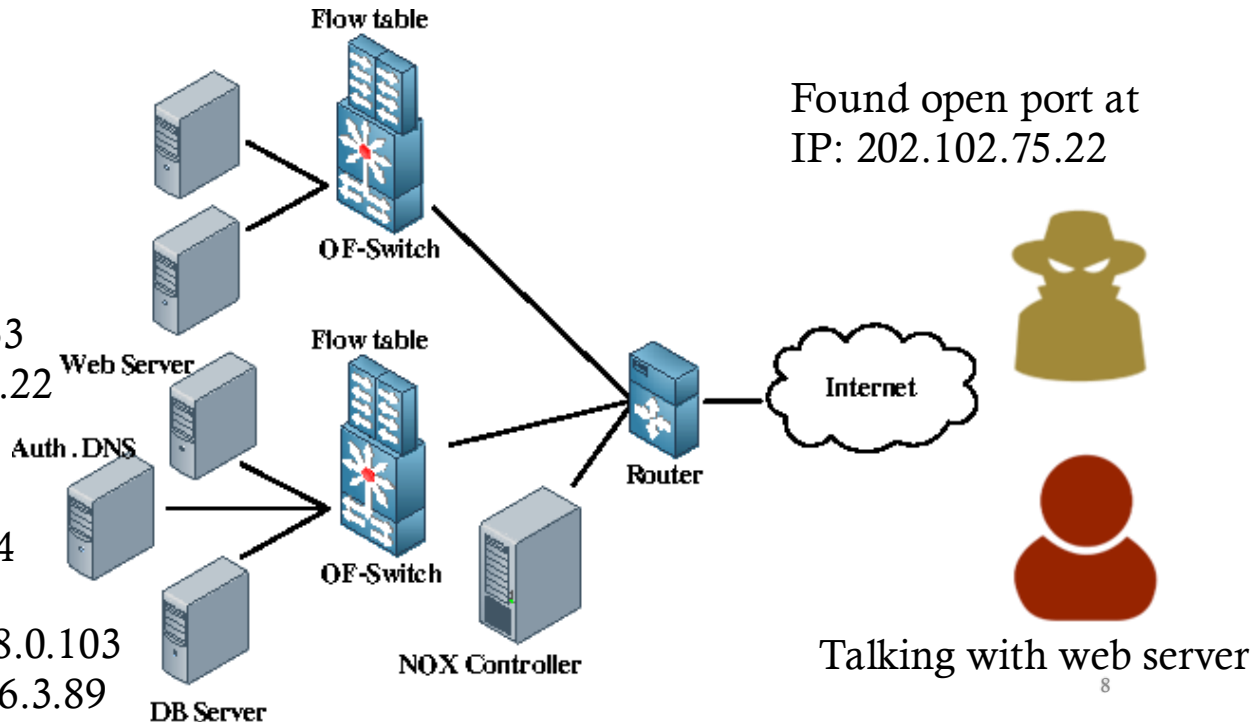
1

Attacker found vulnerabilities in the web server by scanning

rIP: 192.168.0.33
vIP: 202.102.75.22

rIP: 192.168.0.32
vIP: 202.100.64.84

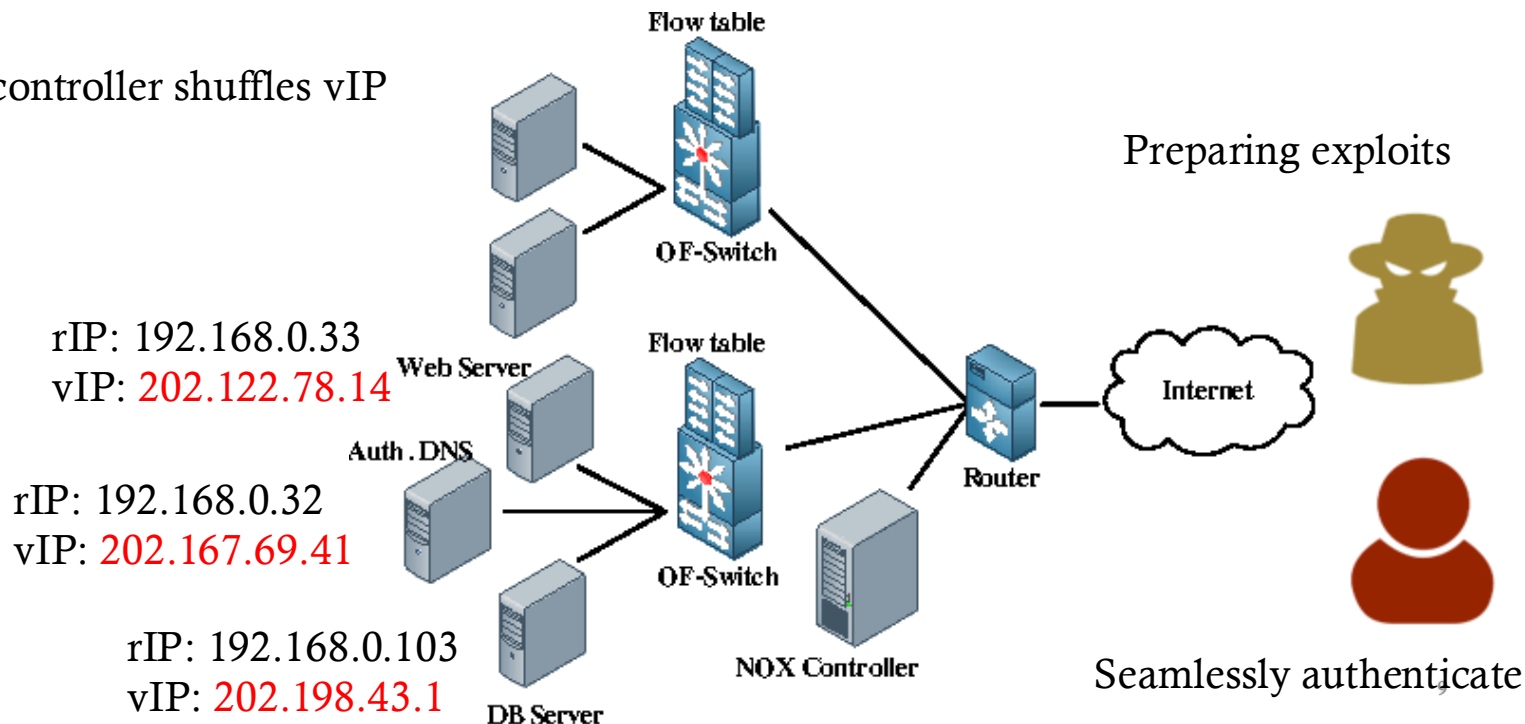
rIP: 192.168.0.103
vIP: 202.116.3.89



MTD Shuffle

2

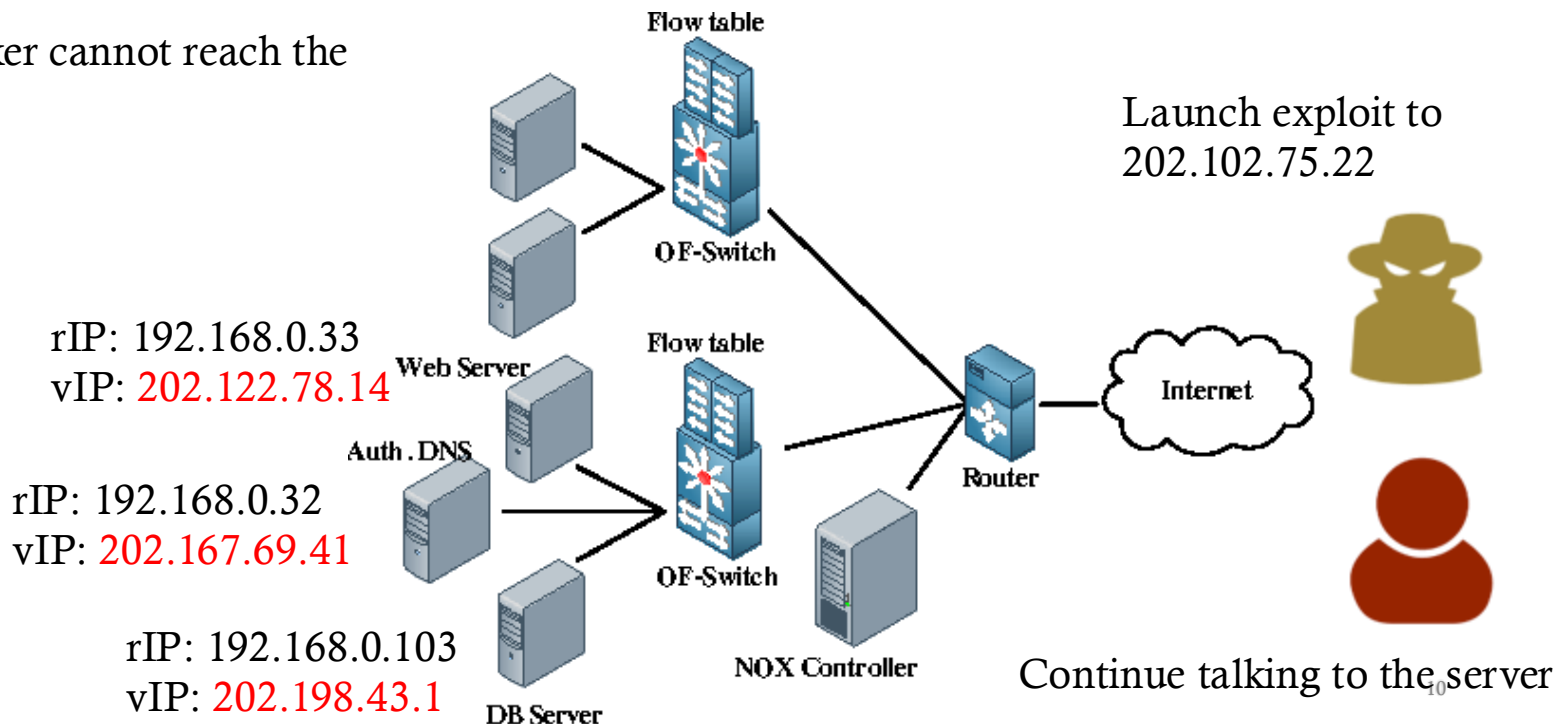
SDN controller shuffles vIP



MTD Shuffle

3

Attacker cannot reach the target



MTD Shuffle

- ◇ What are some issues to consider using shuffle?
 - ◇ Shuffled system state may not be secure
 - ◇ Shuffle may violate security requirements
 - ◇ Shuffle may not be feasible
 - ◇ E.g., cannot satisfy performance requirements
 - ◇ Shuffle may not be applicable
 - ◇ E.g., homogeneous network/system
- ◇ How frequently should we shuffle?
 - ◇ Proactively, reactively, etc.

MTD Diversity

- ◇ Diversity technique provides replacement of an existing component/service, with **different implementation/configuration** that provides the **same functionalities**
- ◇ Existing vulnerabilities may be **replaced** with a new set of vulnerabilities
 - ◇ E.g., replaced the Windows-based computer to Mac

MTD Diversity

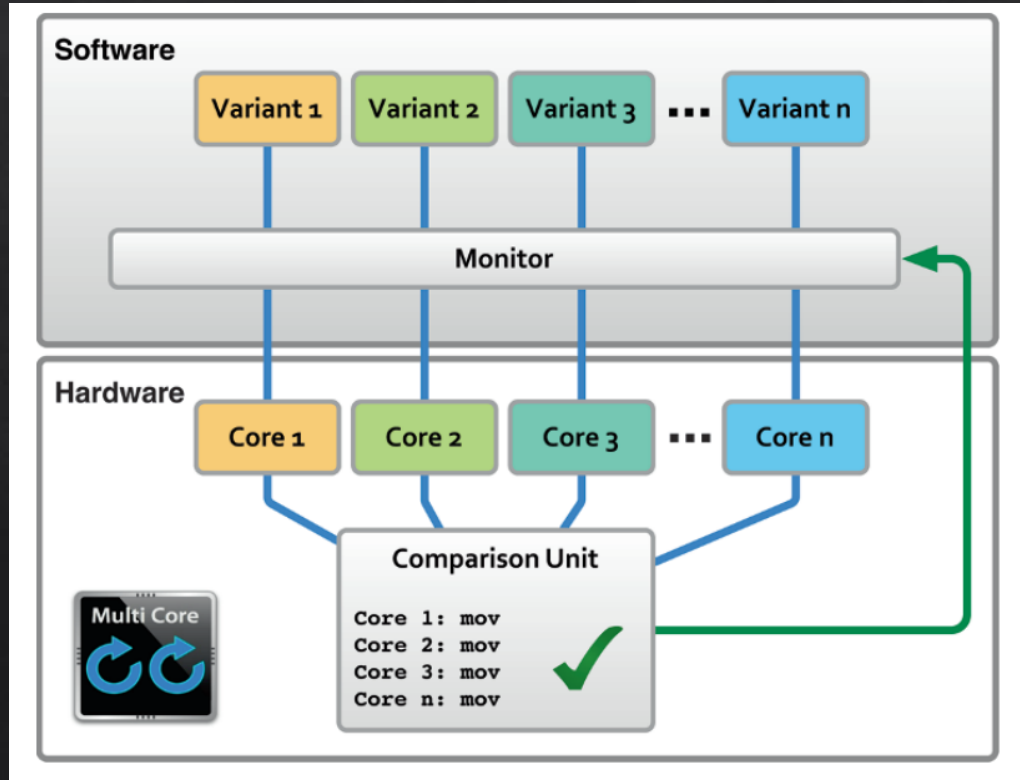
- ◇ Example: Compiler generated software diversity
 - ◇ A version of software has a specific signature/binary
 - ◇ There are number of instructions to carry out the software task
 - ◇ These instructions can be remapped with other (single or combinations) instructions to do the same operations
 - ◇ Simply, $3+5$ is the same as $14-6$

```
movl  %edx, %eax  
xchgl %edx, %eax
```

can be replaced with

```
leal   (%edx), %eax  
xchgl  %edx, %eax
```

MTD Diversity



Variant monitor and comparison units are added to ensure the same operations by the variant software

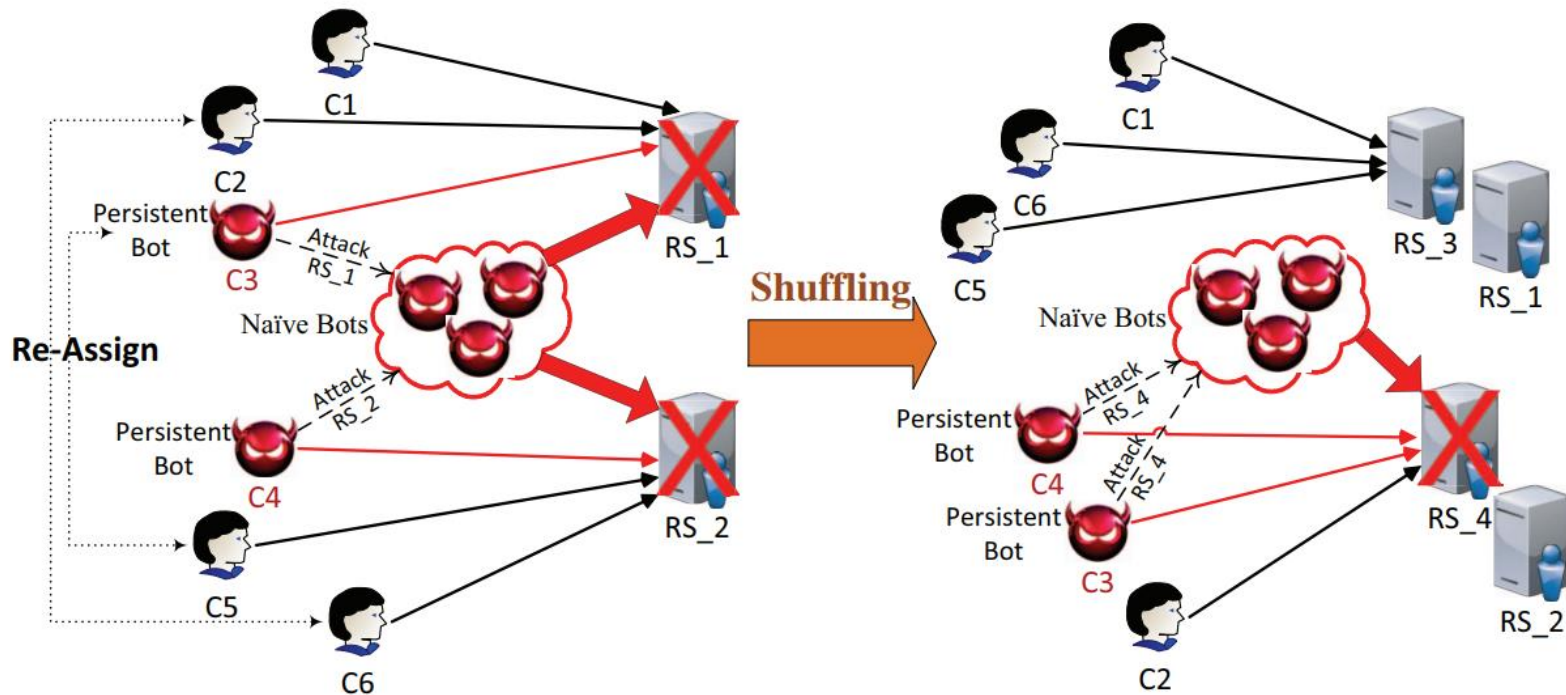
MTD Diversity

- ◇ What are some issues to consider using diversity?
 - ◇ New variant(s) may be more vulnerable
 - ◇ Difficult to generate variants
 - ◇ Cost to implement diversity technique
 - ◇ E.g., OS diversity in servers
 - ◇ Downtime is more significant than shuffle
 - ◇ E.g., rebooting vs SDN flow switching

MTD Redundancy

- ◇ Redundancy aims to ensure the availability of the system
- ◇ Replicas provide resources to handle high usage of the system
- ◇ It can be used in conjunction with other MTD techniques to provide defense against attackers
- ◇ For example DDoS attacks

MTD Redundancy



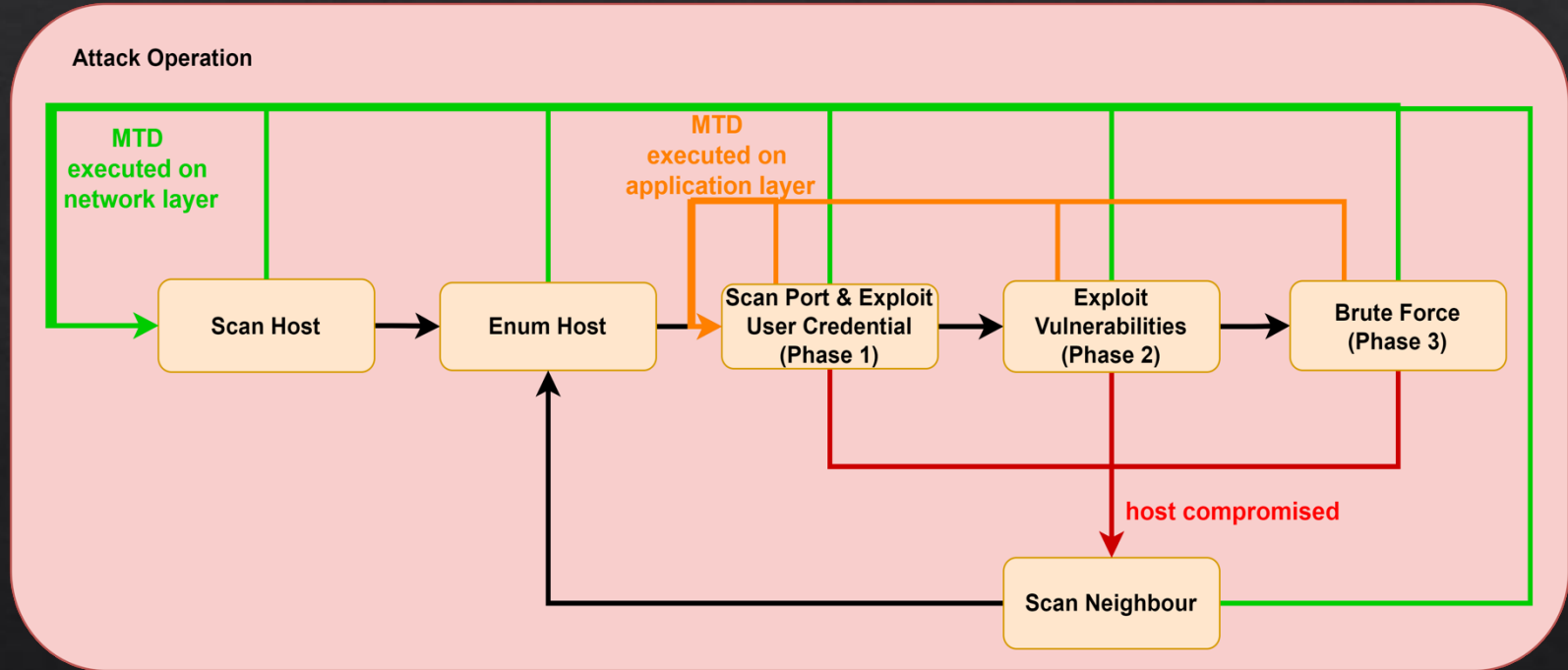
MTD Redundancy

- ◇ By adding redundant servers and shuffling identified bots to the same server, we can ensure the **availability** of the cloud resources to legitimate users
- ◇ Problems with redundancy techniques include
 - ◇ Resource constraints
 - ◇ Ineffective against attacks violating confidentiality and integrity (but could be used for tolerance)
 - ◇ Degrading performance
 - ◇ Increasing redundancy does not proportionally increase availability

What next in MTD?

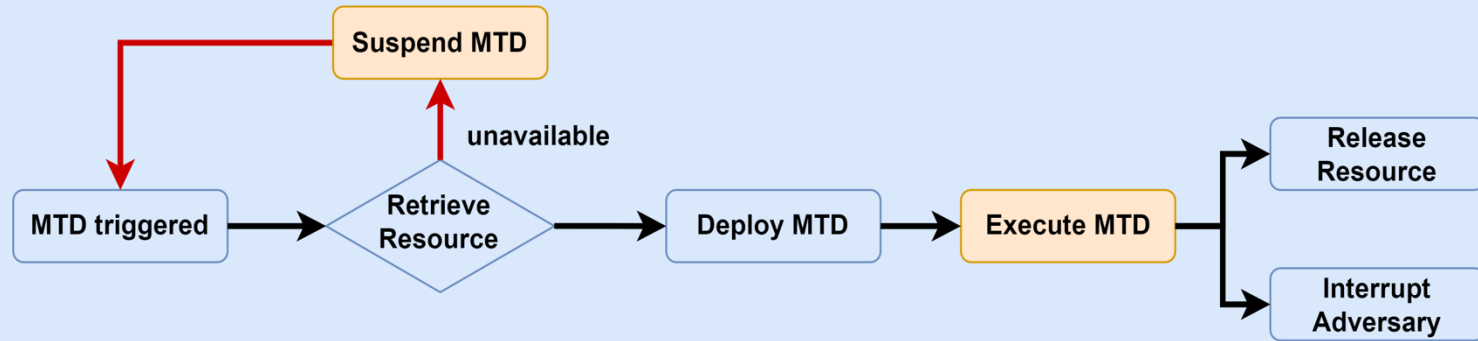
- ◇ Different MTD techniques provide **different** approaches to ensure the **security objectives** of the system
- ◇ Many MTD research has been conducted in the last decade
- ◇ It is difficult to evaluate how effective MTD is in your own system
- ◇ It is uncertain how to optimise MTD deployment scenario
 - ◇ Is one MTD technique enough?
 - ◇ Are two MTD techniques better than one?
 - ◇ Can I randomly deploy MTD techniques?

Attack Operation

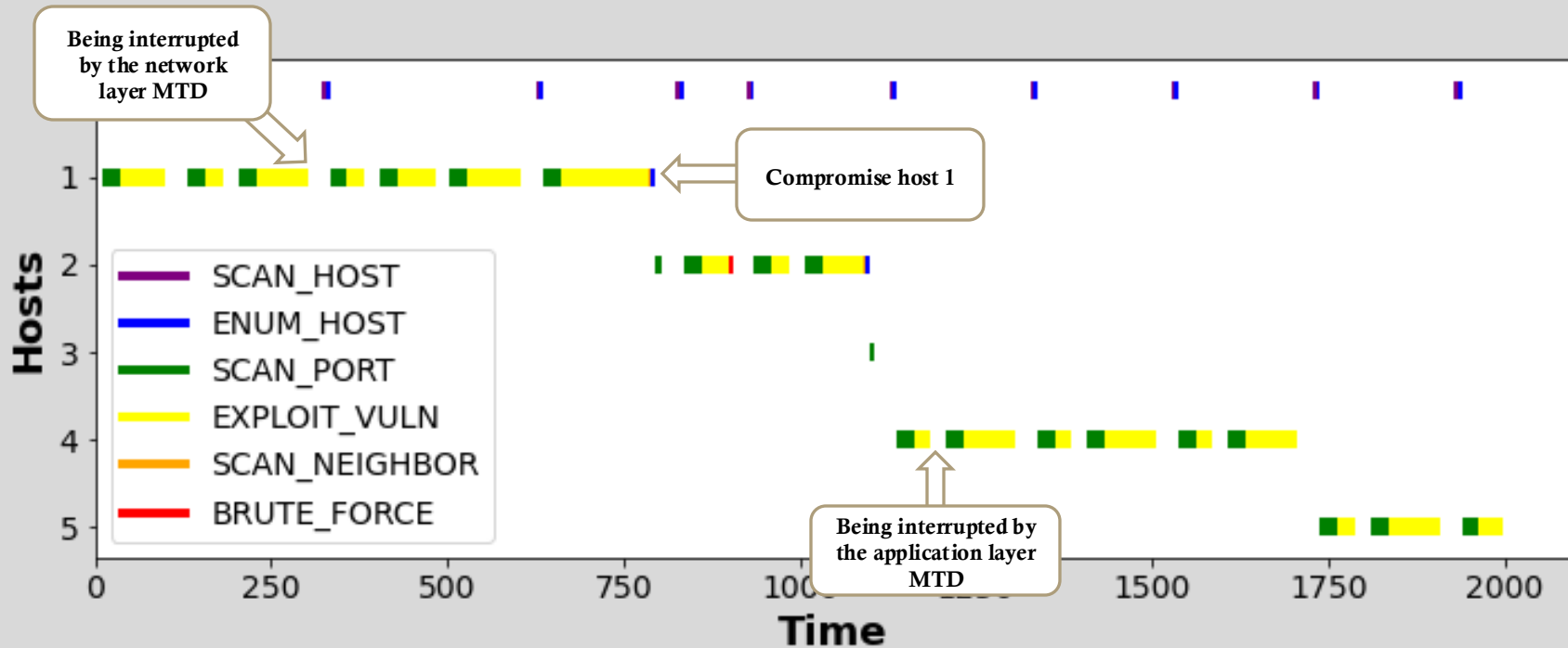


MTD operations

MTD Operation

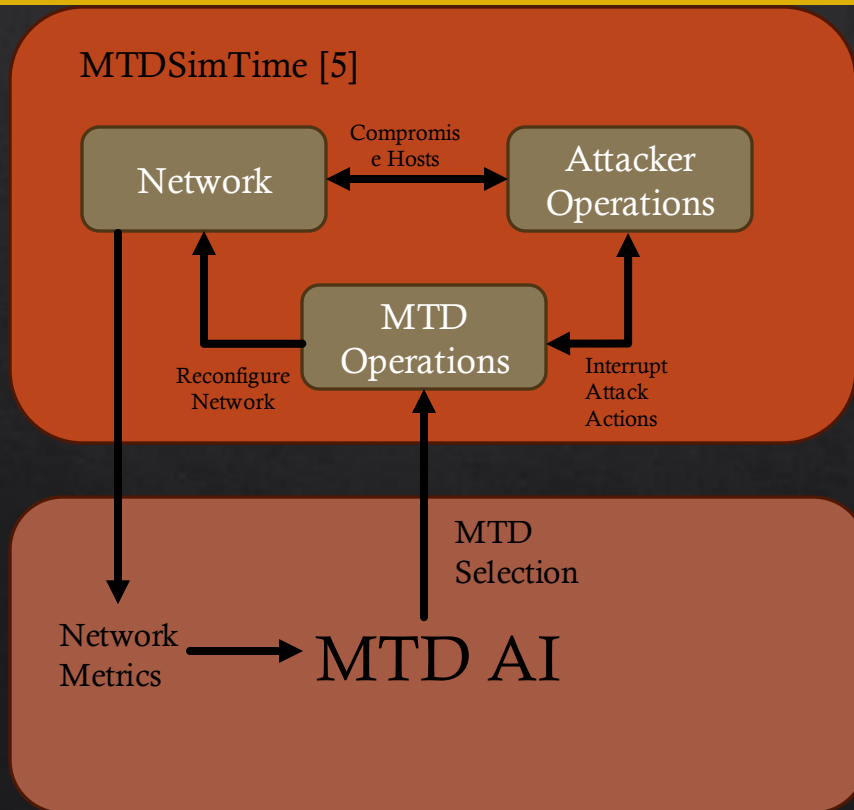


Attack operation visualization



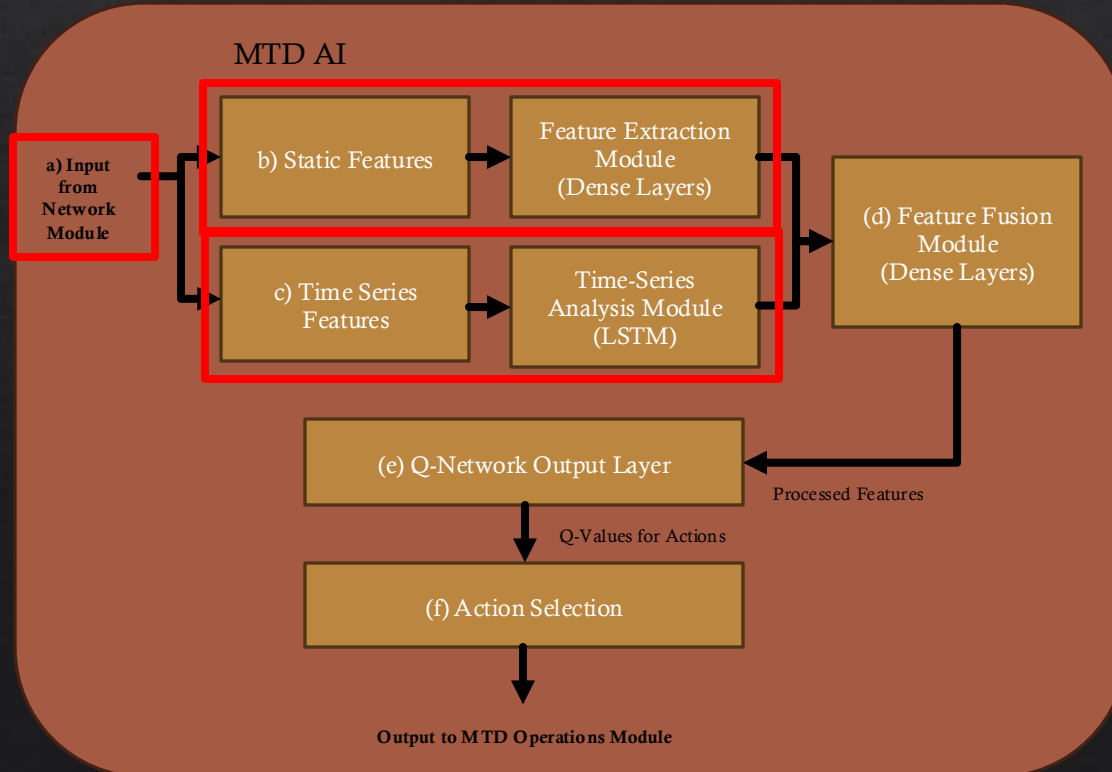
MTD and AI

System Overview



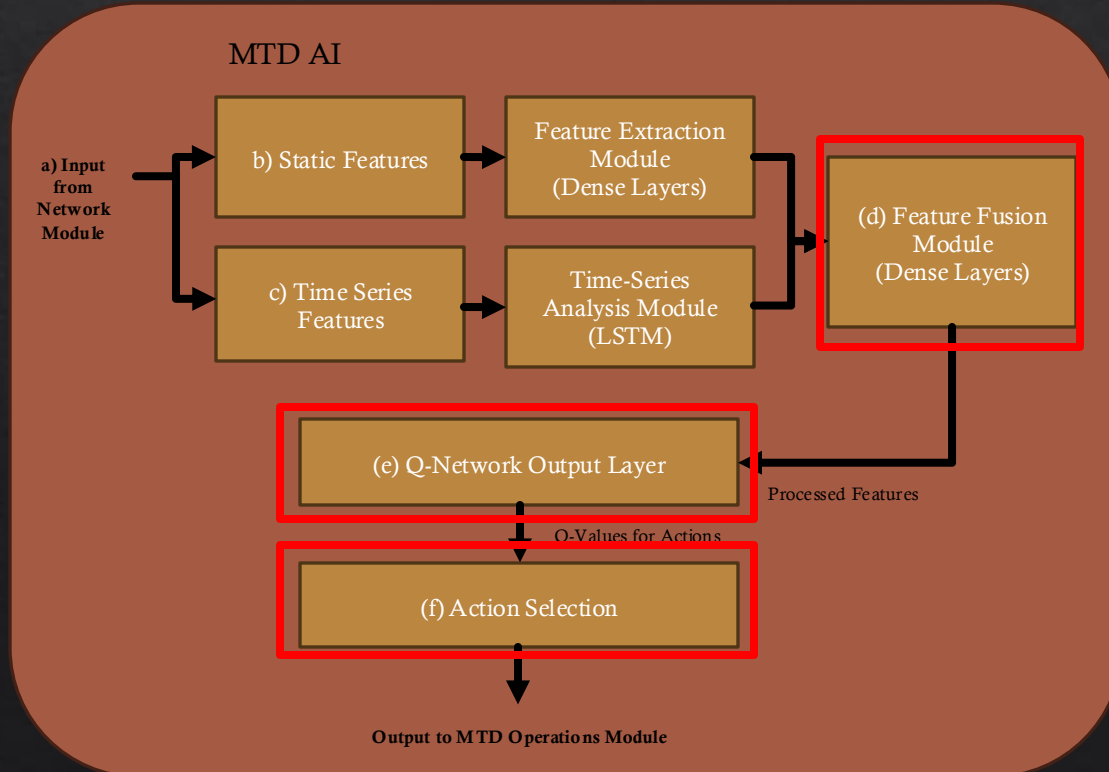
- ❖ MTD Operations module deploys MTD into the network on fixed intervals
- ❖ The reinforcement learning (RL) model receives a stream of data from the network and configures the deployment of MTD techniques in real-time.

Model Architecture



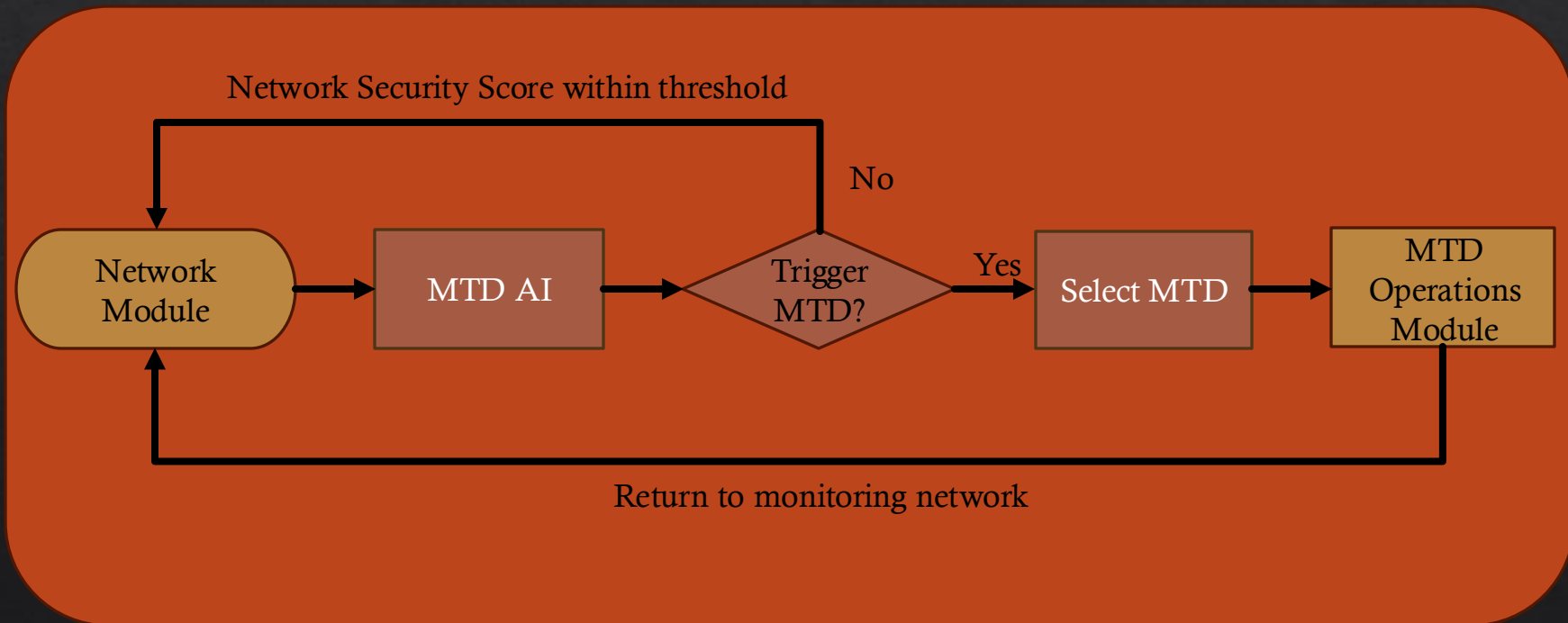
- Static features such as Attack Path Exposure, Host Compromise Ratio and Exposed Vulnerabilities.
- Feature extraction module: Dense Layers -> ReLu -> Dropout
- Temporal features: MTCC, Network Downtime and Time Since last MTD
- LSTM: Captures temporal dependencies in the data

Model Architecture



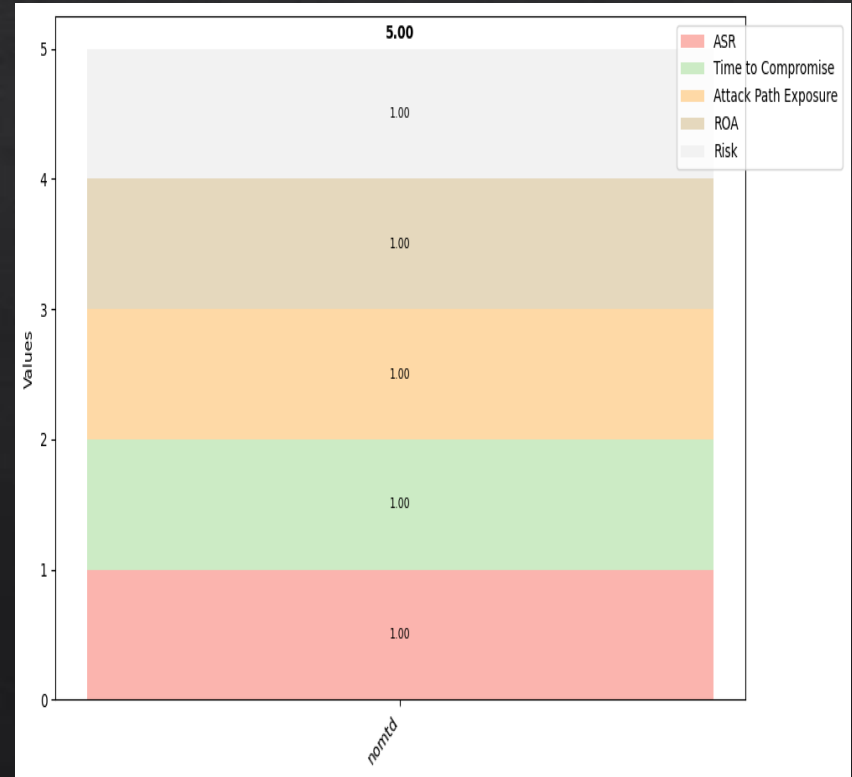
- Feature fusion: Dense layers to integrate the data from the previous 2 modules
- Q-Network Output: Vector of Q-Values
- Action selection tells MTD Module what to do

Model Integration



No MTD Scenario

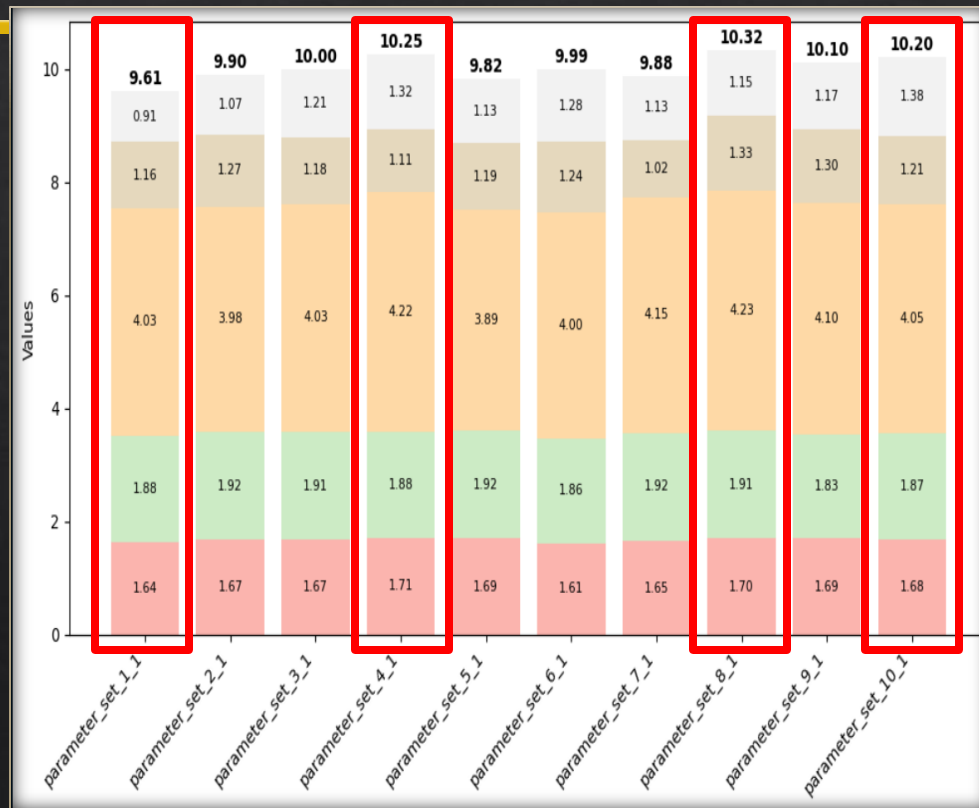
- Base case with no MTD deployed
- All other scenarios will be scaled based on these results



Parameter Sets

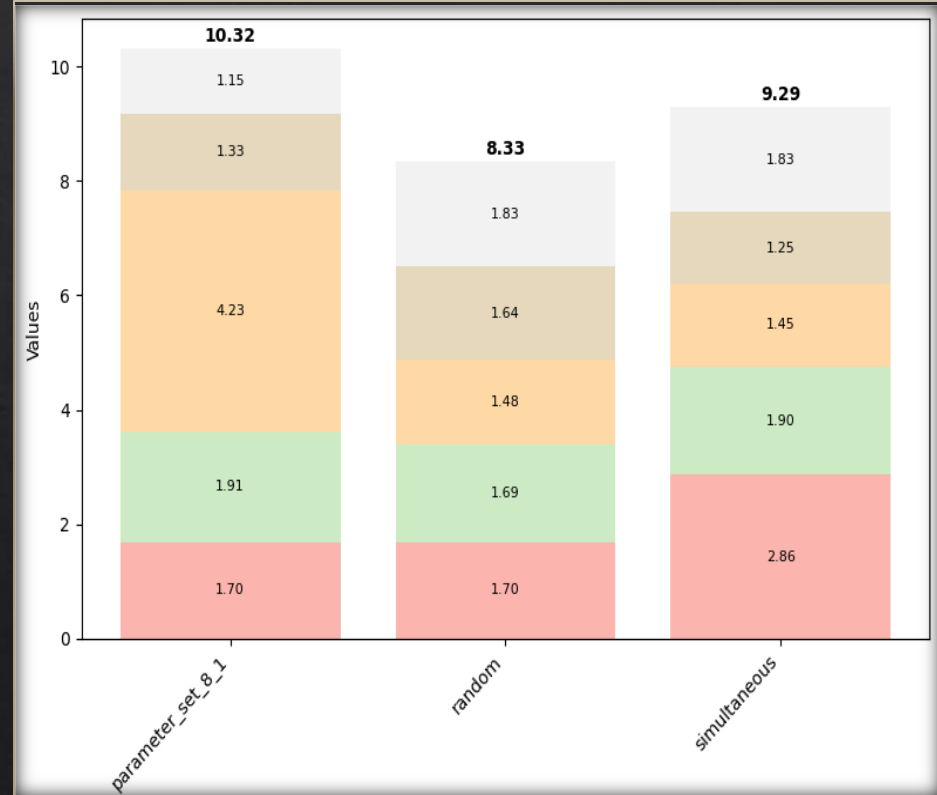
1-10

- Parameter set 8: Balances exploration and exploitation with a large memory buffer and slower epsilon decay, promoting more comprehensive learning.
- Parameter set 4: Starts with high exploration, gradually shifting to exploitation, providing a balanced approach to exploration and exploitation over time.
- Parameter set 10: Optimizes long-term rewards and learning stability by delaying training, collecting more experiences, and training over an extended number of episodes.
- Parameter set 1: Focuses on faster learning with a strong emphasis on short-term rewards and quicker convergence to exploitation.



MTD AI vs Other Schemes

- ◇ Random: Randomly deploys an MTD technique at fixed intervals
- ◇ Simultaneous: Deploys all available MTD techniques at fixed intervals



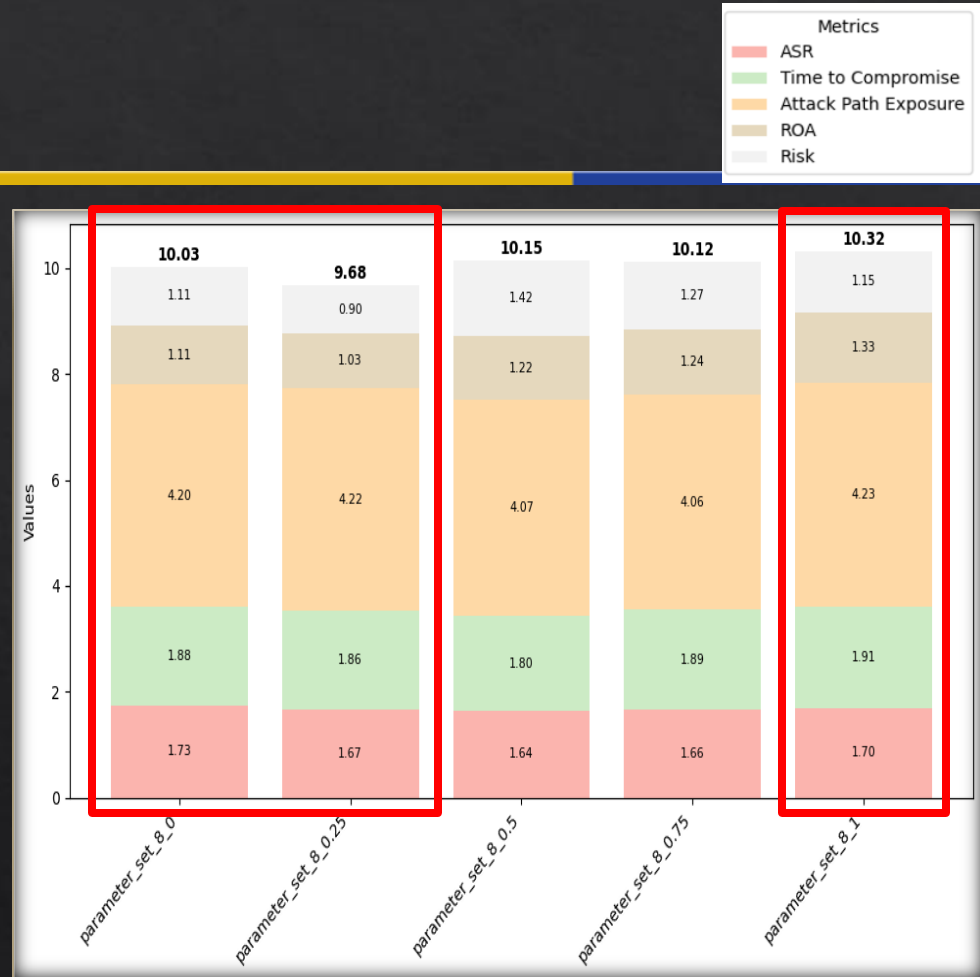
Intrusion Detection Systems

- ◇ Intrusion Detection Systems (IDS) are security solutions designed to monitor network or system activities for malicious actions or policy violations.
- ◇ They monitor unauthorized attacks on the system.
- ◇ Experiments were conducted to determine if the presence of data from the IDS would supplement the ability of the ML model to respond to threats.



Sensitivity Analysis

- ◇ Model with full knowledge of attacker performed the best due to being able to learn how to counter each attack type with MTD
- ◇ Model which received only limited information performed worse than model with no attacker information



Summary

- ◇ MTD is one of proactive security solutions that changes the attack surface continuously
- ◇ Various MTD techniques have been proposed, applicable in multiple system levels
- ◇ It is still difficult to determine how to optimally deploy MTD techniques in given systems and networks
- ◇ Using security modelling and analysis can be useful to understand MTD techniques and their effects in systems and networks
- ◇ Use of AI and ML can assist in configuring MTD deployment strategies

Additional reading

- ◇ Cho, Jin-Hee, Dilli P. Sharma, Hooman Alavizadeh, Seunghyun Yoon, Noam Ben-Asher, Terrence J. Moore, Dong Seong Kim, Hyuk Lim, and Frederica F. Nelson. "Toward proactive, adaptive defense: A survey on moving target defense." *IEEE Communications Surveys & Tutorials* 22, no. 1 (2020): 709-745.